

KEEN Related activities for Eric G. Meyer

Courses taught with EML activities

BME 3113 "Wearable Technology Studio"
EGE 1001 "Fund. of Eng. Design Projects"
BME 5093 "Special Topics: Orthopedics "
BME 3303 "Introduction to Biomechanics"
BME 1002 "Introduction to BME"
BME 4313 "Tissue Mechanics"
BME 3002 "Biomedical Best Practices"
BME 4013 & 4022 "Biomedical Projects"

Publications (Peer-Reviewed)

Meyer EG, Ulrey BL. Sensing Angular Kinematics by Embedding an Open Hardware Design Project into a Required Biomechanics Course. ASEE Annual Conf Proc. 2016.
Rust M, Nasir M, Meyer EG. Using Quantified Self as a Learning Tool to Engage Students in Entrepreneurially Minded Learning and Engineering Design. ASEE Annual Conf Proc. 2016.
Nasir M, Weaver BT, Meyer EG. Project-based Learning in a Forensic Engineering Course. ASEE Annual Conf Proc. 2016.
Meyer EG, Nasir M. Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the "Quantified Self" social movement. ASEE Annual Conf Proc. 2015.
Nasir M, Seta J, Meyer EG. Introducing high school students to biomedical engineering through summer camps. ASEE Annual Conf Proc 2014.
Gerhart A, Meyer EG. Combining discipline-specific "Introduction to Engineering" courses into a single multi-discipline course to foster the entrepreneurial mindset with entrepreneurially minded learning. ASEE Annual Conf Proc 2014

Other Activities

KEEN Shaping Entrepreneurial Engineers 2011 (Tempe, AZ)

KERN Innovative Teaching Program (Southfield, MI)

- Problem Based Learning 2011
- Active and Collaborative Learning 2012

KEEN Entrepreneurially-Minded First Year Experience 2012 (Flint, MI)

KEEN Meeting 2013 (Milwaukee, WI)

Kern Family Foundation KEEN Topical Grant, 2013

- Role: Primary investigator. "Fostering the entrepreneurial mindset through the development of multidisciplinary engineering learning modules based on the "Quantified Self" social movement".
- Role: Contributing Faculty. "Opportunity seeking in a highly regulated product sector: Medical device products from concept to investor pitch".

Thumbworks! Education & Community Day Conference 2014 (Marlette, MI)

- Oral presentation: Medical Leaps & Bounds.

KEEN Winter Conference 2014 (Tempe, AZ)

- Poster: Fostering the entrepreneurial mindset through the development of multidisciplinary engineering learning modules based on the "Quantified Self" social movement.

ASEE Annual Conference 2014 (Indianapolis, IN)

- Poster: Fostering the entrepreneurial mindset through the development of multidisciplinary engineering learning modules based on the “Quantified Self” social movement: Pilot Module Implementation.

Kern Family Foundation KEEN Topical Network Grant, 2014

- Role: Co-PI. “Building a Network for Faculty Collaboration on Entrepreneurship by means of a “Quantified-Self Roadshow”

Lawrence Technological University QS4EML Roadshow 2014 (Southfield, MI)

- Oral presentation: Using Quantified Self for EML in Biomedical Engineering Courses.

KEEN’zine Issue Two, 10/2014

- Article: A Healthy Approach: A course update to encompass KEEN outcomes transforms a biomedical engineering program.

VentureWell Open Conference 2015 (Washington, DC)

- Poster: Entrepreneurship in small doses.

Innovating the Curriculum with Entrepreneurship in Biomedical Engineering 2015 (Southfield, MI)

- Oral presentation: Entrepreneurship at LTU.

KEEN Webinar on 5/7/15

- Quantified Self for Entrepreneurial Minded Learning (QS4EML)

KEEN Blog post 2015

- Article: Biomedical Engineering Innovation – Engaging Our Students.

Bucknell University QS4EML Roadshow 2015 (Lewisburg, PA)

- Oral presentation: Using Quantified Self for EML in Biomedical Engineering Courses.

ASEE Annual Conference 2015 (Seattle, WA)

- Oral presentation: Fostering the entrepreneurial mindset through the development of multidisciplinary learning modules based on the “Quantified Self” social movement.
- Poster: Providing Opportunities for Applied Research and Design-Based Capstone Projects in Biomedical Engineering

Ohio Northern University QS4EML Roadshow 2016 (Ada, OH)

- Oral presentation: Using Quantified Self for EML in Biomedical Engineering Courses.

ASEE NSC Annual Conf. 2016 (Mount Pleasant, MI)

- Oral presentation: Implementation of an Entrepreneurial Minded Learning Module Based on “Quantified Self” to Motivate Engineering Design in an Introductory Biomedical Engineering Course.
- Workshop: Learning Modules Focused on Entrepreneurship in Biomedical Engineering.

Kern Family Foundation KEEN Topical Grant, 2016

- Co-Primary investigator. “Creating a Successful Model for Dissemination and Adoption of Entrepreneurial Mindset in Biomedical Engineering”.

American Society of Engineering Education 2016 (New Orleans, LA)

- Oral presentation titled “*Sensing Angular Kinematics by Embedding an Open-source Electronics Design Project into a Required Biomechanics Course*” in the Biomedical Division (T604). Dr. Meyer was joined by Dr. Brent Ulrey from Western New England University. The talk was very well-received.
- Oral presentation titled “*Using Quantified Self as a Learning Tool to Engage Students in Entrepreneurially Minded Learning and Engineering Design*” with Dr. Meyer as co-authors in the Entrepreneurial Division (T624).
- Dr. Nasir and Dr. Meyer hosted a table during KEEN /ENT division reception with the hands-on kits and made several new contacts with people outside the KEEN network.

Annual Biomedical Engineering Society Conference 2016 (Minneapolis, MN)

- Oral presentation titled *“Fostering Entrepreneurial Mindset in Biomedical Engineering Programs”* in the Biomedical Engineering Education Track. The talk was attended by more than 30 people
- LTU also had a booth at the conference where QS4EML website and brochures were displayed.

BMES – IDEA Conference 2016 (Minneapolis, MN)

- Presented a snapshot of the EML modules and qs4eml.ltu.edu website with audience as well as the plans for BME materials.

KEEN 2017 Annual Conference (Jacksonville, FL)

- Hosted a table with hands-on kits at the Annual KEEN meeting at Networking Session.

BMES Webinar on 4/10/17

- Hosted an international webinar titled *“Entrepreneurship and Innovation in Biomedical Engineering”*.

American Society of Engineering Education 2017 (Columbus, OH)

- Hands-on workshop titled *“Hands-on activities for Prototyping and Ideation in Biomedical Engineering”* in the BME division. The two hour workshop was attended by 11 people.

Kern Family Foundation KEEN Institutional Grant Subaward, 2017

- Role: Co-Primary investigator. *“Interdisciplinary Wearable Technology Studio for Design and Entrepreneurship”*.

BMES 2017 (Phoenix, AZ)

- *“Practicing Design and Prototyping in Biomedical Engineering Programs”* focusing on the newer developments with hands-on kits and prototyping. This won the **BMES Best Poster Award** at the conference.

LTU Workshop on Disruptive Technology for EML on 10/26/17 (Southfield, MI)

- Dr. Michael Rust from Western New England University was hosted for an afternoon workshop with approximately 10 participants from LTU.

Integ8™ Meeting 2017 (Detroit, MI)

- Hosted a table with kits, example projects, and student representatives at an international trade show focused on the 4th Industrial Revolution.

KEEN 2018 Annual Conference (Dallas, TX)

- Workshop titled *“Reinforcing Technical Topics through Hands-on EML Design Modules”*

Detroit Public School Teacher Workshop on 6/1-6/2/18 (Southfield, MI)

- Hands-on workshop titled *“Using Maker Projects To Spark Interest In Stem”*

World Congress of Biomechanics 2018 (Dublin, Ireland)

- Oral presentation on a paper titled *“Building Interactive Wearable Technology Devices to Motivate Hands-on Experience with Biomechanics and Biomedical Engineering Design”*.

A hands-on kit that was used in the ASEE 2017 workshop and KEEN 2018 conference as well as in multiple course modules and other outreach activities. The contents include a combined microcontroller and expansion shield called “Seeduino”, multiple “Grove” sensors (Inertial, light, temperature and sound) and connection cables for communication with the software on the computer / laptop. The part list, suppliers, and costs are details so that anyone interested in making their own can buy the parts directly from the vendors. The teaching and learning modules refer to the kits and can be modified as needed in order to implement a specific activity in classroom.

Dr. Nasir and Dr. Meyer created 6 episodes for “The QS4EML show” focusing in various aspects of the EML mindset and skillset. 4 of these episodes can be found at the channel

<https://www.youtube.com/playlist?list=PLZSch4Qp38sFjlqBa-PiafSQQQQ-kaFDL>

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